26900570 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Skill Enhancement Course (SEC) - MG5SECZGY300, MG5SECZIM300 - FOOD AND WATER QUALITY MANAGEMENT

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Name the adulteration in which papaya seeds mixed with black pepper [U] / [CO1]

   a). Intentional b). Incidental

   c). Occasional d). Temporary

2. The optimal temperature range for mesophilic bacteria is……. [K] / [CO3]

   a). 0-10 °C b). 20-45 °C

   c). 50-70 °C d). -5 to 5 °C

3. Oxidation of fats and oils can result in [K] / [CO4]

   a). Rancidity b). Freezing

   c). Fermentation d). Coagulation

4. What is the main function of a thickening agent? [U] / [CO1]

a). To increase the viscosity of a food

b). To prevent oxidation

c). To provide sweetness

d). To prevent microbial growth

5. The major causative organism for the fruit spoilage is [K] / [CO3]

a). Pencillium

b). Botrytis

c). Pseudomonas

d). All of the above

6. Under strict liability, consumers do not need to prove ..... [U] / [CO2]

a). That they purchased the product

b). Negligence

c). That the product exists

d). The product price

7. Which of the following can help prevent enzymatic browning? [K] / [CO3]

a). Ascorbic acid

b). Sodium chloride only

c). Starch

d). fumaric acid

8. During freezing, most of the water present in food is converted into: [U] / [CO4]

a). Ice crystals

b). Vapour

c). Liquid form

d). Solute

9. Cholera is caused by : [K] / [CO3]

a). *Salmonella typhi*

b). *Staphylococcus*

c). *Klebsiella*

d). *Vibrio cholerae*

10. Which of the following regulates food safety in United States [K] / [CO2]

a). QCI

b). NABCB

c). FAO

d). FDA

11. Which toxic metal is associated with Itai-Itai disease? [U] / [CO1]

a). Cadmium

b). Lead

c). Mercury

d). Arsenic

12. Free fatty acids are formed by the action of [K] / [CO3]

a). Lipase

b). Pectinase

c). Protease d). Invertase

[1x12 = 12]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **3** marks

13. Mention any three factors affecting the rate of drying and dehydration. [U] / [CO4]

14. Describe the common types of spoilage found in milk. [U] / [CO3]

15. What is BIS? Describe its importance in the food industry. [U] / [CO2]

16. Discuss the socioeconomic consequences of food adulteration [U] / [CO1]

17. Describe any three common bacterial identification test. [U] / [CO3]

18. Explain the role of salt in food preservation. [U] / [CO4]

19. What is meat spoilage? Mention any two signs of spoilage in meat. [U] / [CO3]

20. Explain the role of FSSAI in establishing standards for food products [U] / [CO2]

[3x6 = 18]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

21. Explain the purpose of standard setting for food additives. Discuss why maximum permitted levels may be established for particular foods. [U] / [CO1]

22. Describe the causes and results of chemical food spoilage [U] / [CO3]

23. Explain shrink packaging and its role among modern food packaging techniques. [U] / [CO4]

24. Discuss the importance of FSS Regulations 2011 in ensuring food safety and quality. [U] / [CO2]

25. Explain the biochemical changes occuring during fish spoilage. [U] / [CO3]

26. Explain the seven principles of HACCP. [U] / [CO2]

[5x4 = 20]